

Problem

A transfer function is given by

$$T(s) = \frac{100(s + 10)}{s(s + 10)}$$

Sketch the magnitude and phase Bode plots.

Step-by-step solution

Step 1 of 3

Consider the transfer function $T(s)$.

$$\begin{aligned} T(s) &= \frac{100(s + 10)}{s(s + 10)} \\ &= \frac{100}{s} \end{aligned}$$

Substitute $j\omega$ for s in $T(s)$.

$$T(s) = \frac{100}{j\omega} \dots\dots (1)$$

Step 2 of 3

Determine the magnitude of equation 1.

$$\begin{aligned} |T(s)| &= \frac{100}{\sqrt{\omega^2}} \\ &= \frac{100}{\omega} \end{aligned}$$

Calculate the magnitude of the magnitude plot at frequency $\omega = 1 \text{ rad/s}$.

$$\begin{aligned} A &= 20 \log \left| \frac{100}{\omega} \right| \\ &= 20 \log \left| \frac{100}{1} \right| \\ &= 20 \log |100| \\ &= 40 \text{ dB} \end{aligned}$$

Step 3 of 3

Determine the phase of equation 1.

$$\begin{aligned}\phi &= -\tan^{-1}\left[\frac{j\omega}{0}\right] \\ &= -\tan^{-1}[\infty] \\ &= -90^\circ\end{aligned}$$

Draw the magnitude and phase plot as shown in Figure 1.

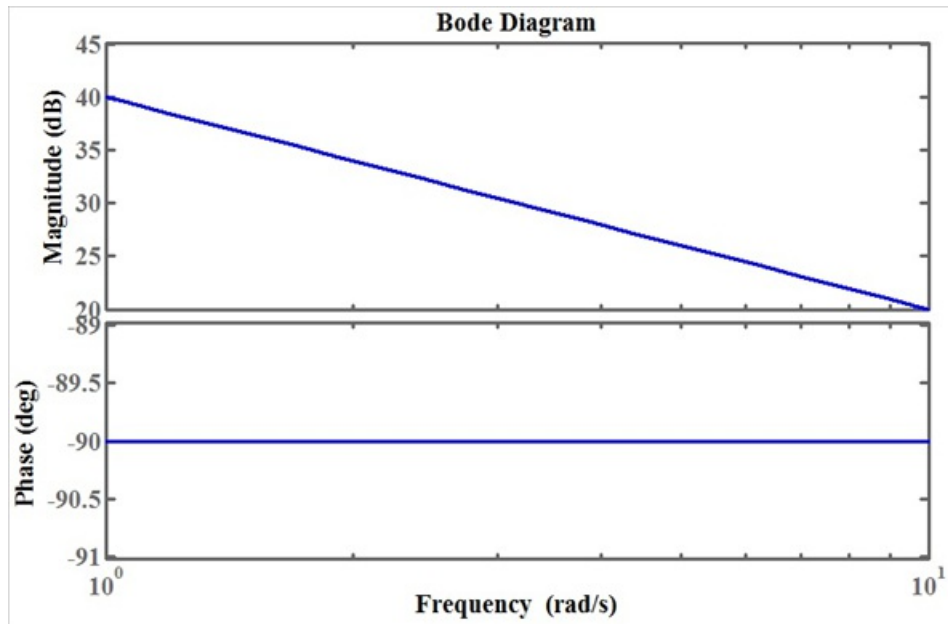


Figure 1

Therefore, the magnitude of $T(s)$ is 40 dB and the phase of $T(s)$ is -90° .